

2. Variational Autoencoder (VAE)

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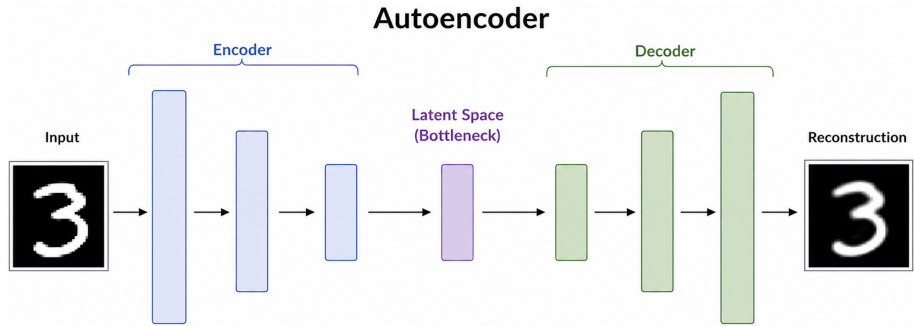
What is an Autoencoder?

- ▶ A neural network that learns to encode data into the latent representation, and reconstruct the original data from that representation.
- ▶ It consists of an encoder, a latent representation, and a decoder.
- ▶ Encoder: maps the input into (generally lower) latent dimensions.
- ▶ Latent representation: Learned feature representation.
- ▶ Decoder: Reconstructs the input from the latent representation.

Autoencoder Reconstruction Objective

- ▶ Input $\mathbf{x}$
- ▶ Latent representation $\mathbf{z} = f_E(\mathbf{x})$
- ▶ Output $\hat{\mathbf{x}} = f_D(\mathbf{z})$
- ▶ Reconstruction loss: $\mathcal{L}_{AE} = ||\mathbf{x} - \hat{\mathbf{x}}||^2$.

Autoencoder Architecture



Autoencoder Applications

- ▶ Dimensionality reduction: similar to PCA
- ▶ Denoising: e.g., noisy images
- ▶ Anomaly detection: reconstruction error
- ▶ Representation learning: learning features

Is an Autoencoder Good at Generation?

Is an Autoencoder Good at Generation?

- ▶ Deterministic encoding and decoding, no uncertainty modeling.
- ▶ Latent space can be irregular and discontinuous.
- ▶ Sampling of the latent space is generally not meaningful.
- ▶ Latent representation interpolation may fail.
- ▶ May memorise the training data.

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- ▶ Decoder only sees valid latent representations from the training data.
- ▶ Randomly sampled latent representations may fall into invalid regions.

Desired Properties of a Generative Latent Space

- ▶ Smoothness
- ▶ Continuity
- ▶ Meaningful sampling
- ▶ Interpolation capacity

What is an Variational Autoencoder?

- ▶ Does not encode a single latent vector.
- ▶ Instead, it encodes a probability distribution.
- ▶ The latent representation $\mathbf{z}_i \sim \mathcal{N}(\mu, \sigma^2)$.
- ▶ The probability density function (pdf) of a normal/Gaussian distribution is $f_{\mathcal{N}}(\mathbf{z}_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(-\frac{(\mathbf{z}_i - \mu)^2}{2\sigma^2}\right)$
- ▶ The latent representation is sampled from the predicted distribution.

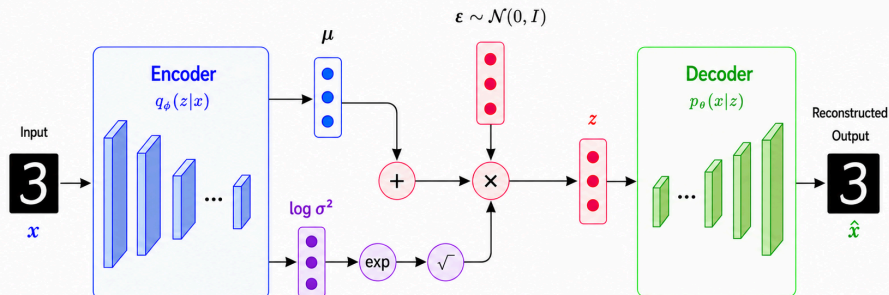
Why the Probabilistic Latent Space?

- ▶ The probabilistic representation spreads information (features) across regions instead of isolated points.
- ▶ Smooth, continuous, and meaningful.
- ▶ Captures uncertainty.
- ▶ Better generalisation, prevents memorisation.

Why Gaussian Distribution?

- ▶ Simple to use.
- ▶ Closed-form solution.
- ▶ Central Limit Theorem: the averages of sufficiently large random samples from any population will be approximately normally distributed, regardless of the population's underlying distribution.

VAE Architecture



VAE Encoder

- ▶ Maps input into a Gaussian distribution.
- ▶ $q_{\phi}(\mathbf{z}_i|\mathbf{x}_i) = \mathcal{N}(\mu, \sigma^2)$
- ▶ Outputs: mean μ , variance σ^2

VAE Latent Space

- ▶ Prior: $p(\mathbf{z}_i) = \mathcal{N}(0, I)$
- ▶ Posterior $q_\phi(\mathbf{z}_i|\mathbf{x}_i)$ is estimated by the encoder.
- ▶ Can we directly sample $\mathbf{z}_i$ from the posterior?

Reparameterisation Trick

- ▶ Direct sampling of $\mathbf{z}_i$ blocks backpropagation.
- ▶ Sample $\epsilon \sim \mathcal{N}(0, I)$.
- ▶ Let $\mathbf{z}_i = \mu + \sigma \odot \epsilon$.
- ▶ Differentiation with respect to μ and σ .
- ▶ Stochasticity is isolated in ϵ .

VAE Decoder

- ▶ Decoder output: $\hat{\mathbf{x}} = f_{\theta}(\mathbf{z})$.
- ▶ Probabilistic interpretation: $p_{\theta}(\mathbf{x}_i|\mathbf{z}_i) = \mathcal{N}(\hat{\mathbf{x}}_i, \sigma^2)$
- ▶ Reconstruction goal: $\hat{\mathbf{x}} \approx \mathbf{x}$.
- ▶ Distribution goal: $q_{\phi}(\mathbf{z}|\mathbf{x}) \approx p(\mathbf{z})$.
- ▶ Trade-off between reconstruction and distribution regularisation.

Kullback-Leibler (KL) divergence

- ▶ It measures how different two probability distributions are.
- ▶ VAE KL divergence term $D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))$
- ▶ Small KL divergence suggests that the distributions are similar.
- ▶ KL divergence is not symmetric.
- ▶ $D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z})) = -\frac{1}{2} \sum_{j=1}^d \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2 \right)$
- ▶ Penalises large mean and variance
- ▶ Penalises deviation from standard Gaussian

VAE Full Objective

- ▶ Learns both reconstruction and a smooth probabilistic latent space.
- ▶ $\mathcal{L}_{\text{VAE}} = -\mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x}|\mathbf{z})] + D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))$
- ▶ Reconstruction loss: $-\mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x}|\mathbf{z})] \propto \|\mathbf{x} - \hat{\mathbf{x}}\|^2$

Evidence Lower Bound (ELBO)

$$\begin{aligned}\log p_{\theta}(x) &= \log \int p_{\theta}(x, z) dz \\&= \log \int q_{\phi}(z|x) \frac{p_{\theta}(x, z)}{q_{\phi}(z|x)} dz \\&= \log \int q_{\phi}(z|x) \frac{p_{\theta}(x|z)p(z)}{q_{\phi}(z|x)} dz \\&\geq \int q_{\phi}(z|x) \log \frac{p_{\theta}(x|z)p(z)}{q_{\phi}(z|x)} dz \\&= \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] + \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p(z)}{q_{\phi}(z|x)} \right] \\&= \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - D_{\text{KL}}(q_{\phi}(z|x) \parallel p(z)) \\&= \mathcal{L}_{ELBO}\end{aligned}$$

VAE Advantages

- ▶ Representation learning
- ▶ Compact embeddings
- ▶ Uncertainty modeling
- ▶ Fast inference
- ▶ Stable training
- ▶ Interpolation
- ▶ Anomaly detection

Summary

- ▶ Autoencoder learns latent representation, but not generative.
- ▶ VAE introduces a probabilistic latent space.
- ▶ VAE encoder predicts mean and variance.
- ▶ VAE latent representations are sampled using reparameterisation trick.
- ▶ VAE cares about both reconstruction and probability distribution.